

I. Introduction

Despite strong evidence of the negative health effects of tobacco use (U.S. Surgeon General 1964) more than 14 billion packs of cigarettes (equivalent to 280 billion cigarettes) were sold in the U.S in 2012 (Orzechowski and Walker, 2013). The Centers for Disease Control (2013) estimate that there are roughly 443,000 smoking related deaths annually and 8.6 million individuals suffer from smoking-related illnesses each year. Excise taxes, levied on manufacturers of tobacco products, are one of the main ways that policymakers can affect tobacco production and consumption.¹ In addition, both state and federal policymakers use tobacco excise taxes to raise revenue. In FY2012, the federal government collected \$15.6 billion in tobacco excise taxes, while the 50 states and D.C. collected \$18.5 billion.

Yet, the impact of excise taxes on consumption and public health is unclear because the taxes are assessed to the manufacturer of tobacco products. Two principal ways that excise taxes could affect tobacco consumption is by changing production and thereby altering the amount or type of tobacco products that are available in the retail market, or by increasing the retail price as a result of pass-through of the taxes. In this paper, I explore both mechanisms in detail. I offer the first analysis of the effect of federal tobacco excise taxes on tobacco *production*. Analyzing production data has several key advantages over individual-level consumption data that are often used to analyze the effect of tobacco excise taxes including less measurement error and detailed data on individual tobacco products. To assess the impact of excise taxes on production, I analyze a large change in the federal excise tax rate enacted in 2009 as part of the Children's Health Insurance Program Reauthorization Act (CHIPRA) using monthly data on tobacco

¹ Bans on smoking are another option available to policymakers. The effect of smoking bans on nonsmokers is unclear. Jerome and Cornaglia (2010) find that smoking bans may increase nonsmokers' exposure to secondhand smoke by forcing smokers into areas where they affect nonsmokers. Carpenter, Postolek and Warman (2011) find that public-place bans on smoking in Canada have large and significant effects on secondhand smoke exposure, especially in bars and restaurants.

production from the U.S. Alcohol and Tobacco Tax and Trade Bureau (TTB) of the U.S. Treasury. I measure the impact of this law change on several categories of tobacco products, including cigarettes, loose tobacco products, cigars and smokeless tobacco. For the price analysis and when measuring the consumer response I focus on cigarettes due to data limitations, although cigarettes account for the vast majority of tobacco consumption. To explore the effect of excise taxes on prices and consumption, I analyze state-year level data on cigarette sales and prices from the Tax Burden on Tobacco (Orzechowski and Walker, 2013).

Exploiting variation in excise tax rates within states over time, I estimate the extent to which excise taxes are passed on to consumers through higher retail prices and the price elasticity of taxable sales of cigarettes.

The results suggest that both producers and consumers respond to tobacco excise taxes in meaningful ways. I find that total production of tobacco decreased by about 180 million pounds on net, the equivalent of 82 billion cigarettes or just over 4 billion packs of cigarettes, in the first 36 months after the enactment of CHIPRA. Decreases in cigarette production drive the decline. I also find strong evidence that manufacturers shift production across substitute products in response to tax differences introduced by CHIPRA. Exploiting variation in cigarette excise tax rates within states over time, I find evidence that both state and federal excise taxes are more than fully passed on to consumers in the form of higher retail prices. Earlier work also finds that tobacco excise taxes are substantively passed on to consumers (Hanson and Sullivan, 2009; Evans, Ringell and Stech, 1999). Using the strong relationship between excise taxes and retail prices in a two-stage least squares (2SLS) estimation, the results suggest that elasticity of taxable sales for cigarettes ranges from -0.27 to -0.43. This finding is similar to the *price elasticity of taxable sales* that Evans, Ringell and Stech (1999) find for an earlier period when both federal

and state excise tax rates were lower. This finding is within the range of estimates of the *price elasticity of demand* for cigarettes (Callison and Kaestner, 2013).

The finding that excise taxes reduce tobacco consumption has important policy implications. Beyond raising revenue, excise tax induced decreases in tobacco consumption may improve public health in several ways, including improving birth outcomes (Evans and Ringel, 1999), lowering tobacco related mortality (Taylor, Hasselblad, Henley, Thurn and Sloan, 2002) and allowing some smokers to overcome time inconsistent preferences helping them to quit (Gruber and Koszegi, 2001, 2004; Gruber and Mullainathan, 2005). Reduced tobacco consumption may also help lower government expenditures on tobacco related medical expenses, estimated to be nearly \$100 billion annually in recent years (Center for Disease Control and Prevention, 2013).

The rest of this paper proceeds as follows. In Section II, I provide background information on tobacco prices and tobacco excise taxes. In Section III, I explore the effect of excise taxes on tobacco production, cigarette prices and cigarette consumption. In Section IV, I conclude.

II. Background

A. Previous Literature

While tobacco excise taxes are levied on manufacturers, the incidence of the tax is likely to be borne in part by consumers through higher retail prices. Economic theory predicts that the ultimate incidence of a tax depends on both the elasticity of supply and the elasticity of demand (Fullerton and Metcalf, 2002). Given the addictive nature of tobacco products the price elasticity of demand is likely to be small so that consumers will bear a relatively large share of the tax.

Consistent with this prediction, Hanson and Sullivan (2009) and Evans, Ringell and Stech (1999) find that tobacco excise taxes are fully passed on to consumers.² Recent work that incorporates search costs (Chernick and Merriman, 2013; Harding, Leibtag and Lovenheim, 2012; Merriman, 2010; Chiou and Muehlegger, 2010; DeCicca, Kenkel, and Liu, 2010; Lovenheim, 2008; Stehr, 2005) suggests that there is substantial heterogeneity in the pass-through of excise taxes to retail prices for consumers who buy cigarettes in bulk, or who smuggle cigarettes across tax borders.³ Unfortunately, the state-level data that I analyze does not allow me to further explore these sources of variation. Instead, the pass through effect that I estimate represents the average effect across all purchases in a given state, including any additional sales that result from cross-border transactions.

Given the findings of substantive pass-through of excise taxes to retail prices, it is expected that tobacco excise taxes will decrease tobacco consumption.⁴ However, the literature does not consistently find this result. Reviewing recent work in this area, Callison and Kaestner (2013) report that the price elasticity of demand for adult smoking ranges from -0.28 to 0.01.⁵ As a further complication, it is difficult to measure the *price elasticity of demand* in survey data, because respondents may not provide accurate information on smoking behavior (Adda and Cornaglia, 2006; Cowling, Johnson, Holbrook, Warnecke, Tang, 2003; Means, Habina, Swan and Jack, 1992). Yet, as survey data provides insight into individual-level tobacco *consumption* it is commonly used in the literature despite this potential drawback. Several studies analyze

² Previous work also finds more than complete pass through for alcohol and gasoline excise taxes (Young and Bielinska-Kwapisz, 2002; Kenkel, 2005; Alm, Senoga and Skidmore, 2009).

³ Results from this work suggests that cross-border smuggling results in a lower pass through of the tax to the retail price and that this type of smuggling is more likely for consumers close to taxing boundaries with meaningful tax differentials. Intuitively, it is easier for consumers closer to these boundaries to seek out lower-taxed cigarettes.

⁴ Fletcher (2012) finds evidence that suggests genetics may be related to tobacco tax responsiveness, with individuals genetically predisposed to tobacco addiction showing less response to taxation.

⁵ Gallet and List (2003) perform a meta-analysis of teen and adult price sensitivity literature and find that the price responsiveness is inversely related to age. In contrast, analyzing data from 1999-2007, Callison and Kaestner (2012) do not find substantive differences in the price elasticity by age.

data from the Current Population Survey (Callison and Kaestner , 2013; Decicca, Kenkel and Liu, 2010; Decicca and McLeod, 2008 and Lovenheim, 2008; Tauras, 2006) and find mixed results.

The literature has taken several steps to address potential problems in survey responses about tobacco consumption. One method is to use scanner-level data (Harding, Leibtan and Lovenheim, 2012; Chiou and Muehlegger, 2010), which is likely to capture a more accurate picture of smoking behavior. Another approach is to use bio-marker data that allows researchers to measure smoking intensity. Evans and Farrelly (1998) and Adda and Cornaglia (2006) suggest that consumers respond along this margin. Yet, as Abrevaya and Puzello (forthcoming) show, in some cases this finding is not robust to alternate econometric techniques so that the impact of excise taxes on intensity is unclear.

A third approach found in the literature is to estimate the *price elasticity of taxable cigarette sales*, which relies on administrative data. To the extent that there is tax avoidance and/or evasion, the elasticity of taxable sales will overestimate the elasticity of demand.

Intuitively, when consumers shift purchases into lower tax jurisdictions the drop in sales in a high tax jurisdiction is offset by increased sales in lower-taxed jurisdictions. Both Evans, Ringell and Stech (1999) and Goolsbee, Lovenheim and Slemrod (2010) take this approach, estimating the *price elasticity of taxable sales*. Evans, Ringell and Stech (1999) find that the elasticity of taxable sales ranges from -0.3 to -0.5, with greater price sensitivity in the long run. Goolsbee, Lovenheim and Slemrod (2010) find that this elasticity has increased in recent years due to the growth in Internet penetration, which allows consumers to obtain lower-taxed cigarettes through on-line transactions. In this paper, I expand this work on the price elasticity of taxable sales in a more recent period that incorporates the CHIPRA law change. While this

measure of price sensitivity may differ from the price elasticity of demand, the price elasticity of taxable sales is a key parameter for federal policymakers looking to generate additional revenue from changes to federal excise tax rates. Recently, both the Obama Administration and members of Congress proposed changes to federal tobacco excise taxes.⁶

B. Tobacco Excise Taxes

Cigarettes account for the majority of tobacco use in the U.S., and a majority of state and federal tobacco excise taxes are from cigarettes. Between 2000 and 2012, cigarettes accounted for more than 92 percent of state tobacco excise taxes and more than 93 percent of federal tobacco excise taxes. In 2012, the average retail price of cigarettes was \$5.64. In that same year, federal excise taxes were \$1.01 per pack and state excise taxes were on average \$1.49 per pack.

Table 1 shows the effect of CHIPRA on tobacco excise tax rates beginning in April, 2009. The cornerstone of the law change was an increase in the excise tax for cigarettes, increasing from \$0.39 per pack to just over \$1.00 per pack. Tax rates were increased on other tobacco products, though not in a uniform manner. Prior to the law change, pipe and roll-your-own (RYO) tobacco were taxed equally at nearly \$1.10 per pound. Following the law change, pipe tobacco increased to \$2.83 per pound while RYO increased to \$24.78 per pound. Pipe and RYO tobacco are likely to be substitutes. The Internal Revenue Code definitions imply that the transformation of RYO to pipe tobacco may be as easy as relabeling RYO as pipe tobacco.⁷

⁶ In the FY2014 Budget, the Obama Administration proposed to increase (and index) tobacco taxes by roughly 95 percent (U.S. Office of Management and Budget, 2013). Senator Richard Durbin proposed the “Tobacco Tax Equity Act of 2013” (S. 194) that would harmonize tax rates across certain product categories.

⁷ According to the Internal Revenue Code, pipe tobacco is any tobacco which, because of its appearance, type, packaging, or labeling, is suitable for use and likely to be offered to, or purchased by, consumers as tobacco to be smoked in a pipe. RYO is any tobacco which, because of its appearance, type, packaging, or labeling, is suitable for use and likely to be offered to, or purchased by, consumers as tobacco for making cigarettes or cigars, or for use as wrappers thereof.

CHIPRA also introduced a large tax differential across small and large cigars. As shown in Table 2, CHIPRA increased the tax on small cigars (defined as having 3 pounds of tobacco or less per 1,000 cigars) from less than \$2.00 per 1,000 cigars to over \$50 per 1,000 cigars. By comparison, the tax increase for large cigars (defined as having more than 3 pounds of tobacco per 1,000 cigars) was smaller, especially for lower-priced cigars. For large cigars retailing at \$50 per 1,000 the federal excise tax increased from roughly \$10 per 1,000 to just over \$26 per 1,000.

III. Empirical Analysis

A. The Impact of CHIPRA on Tobacco Production

To assess the impact of CHIPRA on tobacco production, I first estimate the counterfactual levels of tobacco production that would have been expected absent the CHIPRA law change. I compare the actual levels of production to the counterfactual levels of production and I attribute differences in production to the CHIPRA law change. The key identifying assumption of this approach is that the *relationship between production and the included covariates would have remained constant in the post-CHIPRA period.*

I use monthly production data from the U.S. Treasury Alcohol and Tobacco Tax and Trade Bureau (TTB) to measure tobacco production. These data include all U.S. production as well as all tobacco products that are imported to the U.S. by both domestic and overseas producers. Column (1) of Table 2 gives the average monthly production from January 2000 through October 2013. Columns (3) and (5) give average monthly production levels in the pre-CHIPRA period (April 2002 through March 2009) and the post-CHIPRA period (April 2009

through March 2012).⁸ As shown in Column (7), there are many significant differences in tobacco production across the pre- and post-CHIPRA periods. Yet, it is not clear if these differences would have occurred in the absence of the CHIPRA law change. By estimating a counterfactual level of production for tobacco products, I can provide insight into whether the observed changes in the post-CHIPRA production levels *differ from the expected levels given the patterns in the pre-CHIPRA period*.

To estimate the counterfactual levels of production, I parameterize the relationship between seasonality, a time trend and state-level cigarette excise rates in a period *prior to the CHIPRA law change* using Equation (1).

$$(1) \quad Production_{jt} = \sum_{t=2}^{12} \delta_t Month_t + \beta_j Trend_t + \lambda StateExciseTax_t + \varepsilon_{jt}$$

In Equation (1), j denotes the type of tobacco product, including cigarettes (in billions of cigarettes), pipe tobacco (in millions of pounds), RYO tobacco (in millions of pounds), small cigars (in millions of cigars), large cigars (in millions of cigars), chewing tobacco (in millions of pounds) and snuff (in millions of pounds) and t denotes months. To account for seasonal variation in production, I include month dummies, and I also analyze symmetric sets of months in the pre- and post-CHIPRA periods. I estimate Equation (1) on 84 months (April 2002 to March 2009) of data prior to the law change, during a time when federal excise tax rates were constant. I then use these estimates on 36 months following the law change running from April 2009 through March 2012.⁹ The federal tax rates for each product appear in Columns (2) and (3) of Table 1. (As a robustness check, I also analyze periods that omit months immediately

⁸ Note that the pre-CHIPRA period is chosen so that a 12-month period runs from April through March to coincide with the CHIPRA start date in April. The federal excise tax rates are constant during all months of the pre-CHIPRA period.

⁹ Results for alternate sample periods that omit months immediately preceding and following the CHIPRA law change are similar. I report these results in Appendix Table A3.

preceding and following enactment of CHIPRA and the results are similar.) To further account for trends in tobacco production, I include a linear time trend (the results are similar when I include higher order trend terms). I also include the consumption weighted state cigarette excise tax rate in each year for all products, both for cigarettes and for non-cigarette products, though the results are similar if I use population-based weights.¹⁰ Due to differences in the base of taxation for non-cigarette tobacco products, I do not include separate effective tax rates for tobacco products other than cigarettes. However, state cigarette tax rates are strongly correlated with tax rates on non-cigarette products.¹¹

Panel A of Table 3 shows the in sample performance of Equation (1). Generally, the model fits the data well. The R^2 for tobacco products except pipe tobacco ranges from 0.73 to 0.93. In the case of pipe tobacco, the R^2 is 0.40. In all cases, the F-test for the included covariates can reject joint non-significance at the 0.01 level. As a further check on the validity of the model, I compare the predicted counterfactual levels *within the pre-law change period*. To do this, I estimate the model on the first 72 months of the pre-sample period, leaving a forecast period of 12 months. Both the mean absolute error and the mean absolute percent error imply that the relatively simple specification does a good job of establishing counterfactual levels of production.¹² Panel B of Table 3 shows these measures of forecast accuracy for each tobacco product. The mean absolute percent error ranges from 4 to 11 percent. The forecast error is smallest for cigarettes, which account for the majority of tobacco use.

¹⁰ Results from a Granger causality test suggest that the consumption weighted state cigarette excise tax rate is not substantively or significantly related to lags of cigarette production. This suggests that the current excise tax rate does not depend on lagged production. These results appear in Appendix Table A1.

¹¹ Appendix Table A2 shows state excise tax rates for pipe, RYO and cigars in 2010, as well as changes to these taxes between 2008 and 2010.

¹² This interpretation is qualitatively similar using other splits of the pre-law change period, including fitting the model to 48 or to 60 months.

Using the parameter estimates from Equation (1), I predict the counterfactual level of production for each tobacco product j in the post-CHIPRA period (April 2009 through March 2012). Figure 1A shows the in sample predicted and actual production for cigarettes, including the 95th percent confidence interval for the predicted counterfactual level of production in the 84 months immediately preceding the CHIPRA law change. Actual production generally falls within or close to the 95th percent confidence interval. Figure 1B shows the predicted and actual levels of production in the first 36 months of the post-CHIPRA period. The counterfactual level is generally above the actual level of production in the post-CHIPRA period, implying that cigarette production fell compared to the predicted levels. Figure 1C plots the differences in predicted and actual production for each month in the post-CHIPRA period, and denotes the statistical significance of the difference. The predicted difference is large and negative for most months, and significantly different from zero at the 0.05 (0.10) level for 24 months (27 months). The pattern in Figure 1C suggests that cigarette production fell by 95 billion cigarettes (equivalent to 4.75 billion packs) or roughly 10 percent in the 3 years after CHIPRA relative to the counterfactual level of production.

Not all forms of tobacco production fell relative to the counterfactual level in the first 36 months of the post-CHIPRA period. In two cases, production appears to have switched away from high-tax products into lower-taxed substitute products. Figure 2 shows predicted and actual pipe tobacco production in both the pre- and post-CHIPRA periods. The model generally fits the data closely in the months prior to CHIPRA enactment. However, this close fit changes in the post-CHIPRA period, when the counterfactual level of production is substantively and significantly different from actual production in each of the first 36 months of the post-CHIPRA period. Figure 3, which shows predicted and actual RYO production, suggests one explanation

for the increase in pipe tobacco production. RYO production fell significantly below the counterfactual level of production in the post-CHIPRA period. The magnitude of the decrease in RYO tobacco production (roughly 56 million pounds) was less than the increase in pipe-tobacco production (68 million pounds), suggesting a net increase in loose tobacco production.

Compared to the counterfactual levels in the first three years after the law change, production of pipe tobacco increased by more than 400 percent while production of RYO tobacco fell by more than 74 percent.

Production changes across cigar categories are another example of a shift in production towards a lower-tax product away from a higher-tax substitute product. Figure 4 shows the predicted and actual levels of large cigar production. Production is substantively and significantly higher than the counterfactual level in each of the first 36 months of the post-CHIPRA period. Production was roughly 17 billion cigars more than the counterfactual level, an increase of 86 percent, in the three years following CHIPRA. Much of this cigar production appears to have switched away from small cigars, which fell significantly relative to the counterfactual level of production in the first three years of the post-CHIPRA period. Figure 5 shows predicted and actual small cigar production. Small cigar production fell by nearly 19 billion cigars in the 36 months following CHIPRA relative to the counterfactual level, a decrease of roughly 86 percent. On net, the results suggest that total cigar production decreased, but the weight of cigars produced may have increased.¹³ However, it is not clear that cigar producers are adding tobacco to otherwise small cigars to meet the large cigar weight threshold. A report in

¹³ Although TTB does not record the amount of tobacco in each cigar, small cigars must have 3 pounds or less per 1,000 and large cigars must have more than 3 pounds per 1,000. If the ratio of tobacco in large cigars relative to small cigars is more than 1.12, then the estimated changes in cigar production will also be an increase in the amount of tobacco.

Bloomberg News (Edney, March 1, 2013) reports that cigar producers are adding clay as well as additional tobacco to meet the large cigar weight limit.

In aggregate, the results suggest that CHIPRA lowered tobacco production by more than 180 million pounds, or roughly 6 percent, in the three years following enactment, compared to the counterfactual level of production.¹⁴ Figure 6 shows the estimated total effect of CHIPRA on tobacco production, taking into account predicted changes in cigarette, pipe tobacco, RYO tobacco, small cigars, large cigars, snuff and chewing tobacco.¹⁵ Decreases in cigarette production, RYO production and small cigar production were partially offset by increases in large cigar production and pipe tobacco production.¹⁶ To put this decrease in perspective, a 180 million pound decrease is equivalent to about 82 billion cigarettes. Given that there are roughly 50 million smokers in the U.S., this suggests a reduction of nearly 30 packs per smoker per year in the first three years of CHIPRA. Shaw, Mitchell and Dorling (2000) report that smoking one cigarette decreases life expectancy by 11 minutes, on average. Together, these results suggest that the estimated decrease in tobacco production from CHIPRA in just the first three years of the bill's passage (assuming all production is consumed) will extend the average lifespan of each smoker in the U.S. by more than 13 days.¹⁷

¹⁴ To generate pound equivalents, I assume that there is 2.2 pounds per 1,000 cigarettes, 2.3 pounds per 1,000 small cigars, and 4.0 pounds per 1,000 large cigars.

¹⁵ The impact of CHIPRA on chewing tobacco and snuff tobacco is unclear. Federal excise rates increased for both of these products following CHIPRA. Appendix Figures A1-A2 show the out of sample results for chewing tobacco and snuff.

¹⁶ Appendix Figure A3 shows predicted changes in pipe and RYO tobacco production and Appendix Figure A4 shows predicted changes in cigar production.

¹⁷ There is roughly 2.2 lbs. of tobacco per 1,000 cigarettes. There are 1,800 cigarettes in 90 packs of cigarettes (30 packs/year for three years). This implies an effect of 19,800 minutes per smoker, or 13.75 days.

B. The Impact of Excise Taxes on Cigarette Prices

To measure the effect of cigarette excise taxes on the retail price of cigarettes, I use data from the Tax Burden on Tobacco (Orzechowski and Walker, 2013). These data include annual state-level information on excise tax rates and cigarette sales for the 50 states and D.C. Due to data limitations, I focus only on the case of cigarettes, though cigarette use accounts for the vast majority of all tobacco consumption. I analyze price changes that include generic cigarettes as well as price changes that exclude generic cigarettes. As described in Evans, Ringel and Stech (1999), generic cigarettes are an important part of the cigarette market. Generic cigarettes are discounted by as much as 40 percent compared to name-brand cigarettes, and their market share has exceeded one-third in recent years. Table 4 shows the average prices and average tax rates for the years 2000-2012. As expected, prices excluding generic cigarettes are higher in this period, averaging \$5.43 per pack compared to \$5.08 per pack for prices including generic cigarettes (\$2012).

To measure the extent to which excise taxes are passed on to consumers in the form of higher retail prices, I estimate Equation (2) on annual state-level data from 2000 to 2012.

$$(2) \quad \text{Price}_{it} = \beta_1 \text{ExciseTax}_{it} + \sum_{i=2}^{51} \delta_i \text{State}_i + \beta_2 f(\text{Time}_t) + \beta_3 X_{it} + \varepsilon_{it}$$

In Equation (2), the price and excise tax rate are in constant cents (\$2012) and the subscripts i and t denote state and year. The key variable of interest in Equation (2) is the excise tax. I estimate specifications that use the combined (state plus federal) tax rate, as well as specifications that include these terms separately.

By including state fixed effects in Equation (2), variation is limited to changes within states over time.¹⁸ I also control flexibly for time effects in several ways. When using the combined tax rate, I am able to include year indicator variables. Given the limited amount of time-series variation in federal excise tax rates, specifications with separate state and federal tax variables cannot include year dummy variables. For these specifications, I include either a common linear time trend across states or state-specific linear time trends (the results are similar when I include higher order trend terms). In X , I control for time-varying state-level characteristics including state unemployment rate and state population.¹⁹ To determine the national level of pass-through, I weight each state-year level observation by the number of packs of cigarettes sold in that state in that year, though the results are similar when I use population-based weights and when I do not use weights.²⁰

A key identifying assumption of this approach is that the tax rate is exogenous to the current retail price. One way that this assumption may be violated is if prices are serially correlated and policymakers select future excise tax rates based on current retail prices. Using Granger causality tests, I do not find evidence that lagged cigarette prices (or lagged sales of cigarettes) in a state are significantly or substantively related to current state excise tax rates. The results for the combined tax rate appear in Appendix Table A4.

The results suggest that excise taxes are more-than fully passed on to consumers in the form of higher retail prices. Table 5 shows the regression results. More-than complete pass through holds for both prices including generic cigarettes as well as prices excluding generic

¹⁸ The inclusion of state fixed effects guards against the potential bias of anti-smoking sentiment as an omitted variable, as discussed in Keeler, Hu, Manning and Sung (2001).

¹⁹ I use data from the Bureau of Labor Statistics, Local Area Unemployment Statistics tables for state populations and state unemployment rates.

²⁰ Due to data limitations, I use the number of packs sold including generic cigarettes in both cases. This approach assumes that the share of generic cigarettes sold is comparable across states.

cigarettes, across all time controls. The estimated pass-through of the combined excise tax rate is 1.05 in Column (1) that includes generic cigarettes and year fixed effects. This implies that the retail price of cigarettes increases by 1.05 cents in response to a 1.00 cent increase in the combined excise tax rate.²¹ The results that include generic cigarettes are slightly smaller, compared to the results that exclude generic cigarettes, though the difference across samples is not significant. (Unfortunately, I do not have data on generic prices separately.) The results are similar for specifications that include a common linear time trend. In Column (2) that includes generic cigarettes, the estimated pass through is 1.15.

The results in Column (3) suggest that there is heterogeneity in the amount of pass through across state (1.10) and federal excise taxes (1.53). The pattern of relatively larger pass-through of federal excise tax rates also holds in Column (5), which uses state-specific linear time trends as time controls. Relatively higher pass through of federal excise taxes is consistent with work by Barnett, Keller and Hu (1995) who predict this result based on market power of cigarette firms and cross-state competition for cigarettes. More generally, greater than one-to-one pass through is consistent with an oligopolistic market structure (Barnett, Keller and Hu, 1995; Harris, 1987).²² Evans, Ringel and Stech (1999), and DeCicca, Kenkel and Liu (2010) also find evidence of substantive pass through of excise taxes to retail prices.

From a public health perspective, the pass-through of the excise tax to the retail price may be more effective at curbing consumption than less obvious or less salient taxes. Chetty, Looney and Kroft (2009) find evidence that consumers respond more to the pass-through of

²¹ Using quantile regressions, I find evidence that the distribution of the pass through effect is bound relatively tightly around full pass through. Although not shown in Table 5, the estimated pass through at the 25th percentile (75th percentile) is 1.033 with a standard error of 0.016 (1.070 with a standard error of 0.008).

²² Gruber (2001) reports that the industry is dominated by four large firms. He reports that in 1998 the market share of these four firms was Phillip Morris (49 percent of market), R.J. Reynolds (25 percent of market), Brown and Williamson (16 percent of market) and Lorillard (9 percent of market).

excise taxes on beer that are included in the retail price than to retail sales taxes that are not included in the posted price but that are added at the register. In the case of cigarettes, Goldin and Homonoff (2013) find suggestive evidence that the relative difference in responsiveness to sales taxes versus excise taxes differs across income groups. They find that low-income taxpayers are less responsive to excise taxes compared to high-income taxpayers.

C. The Price Elasticity of Cigarettes

A key problem in estimating the price elasticity for cigarettes is isolating variation in prices that is plausibly exogenous to consumption. Building on the results from the previous section, I use excise taxes as instrumental variables for cigarette prices to estimate the *price elasticity of taxable sales of cigarettes*.²³ To estimate the price elasticity of taxable cigarette sales, I focus on sales and prices that include generic cigarettes (I do not have separate sales data that exclude generic cigarettes) using Equation (3).

$$(3) \quad \ln(\text{Cigarettes}_{it}) = \beta_1 \ln(\text{Price}_{it}) + \sum_{i=2}^{51} \delta_i \text{State}_i + \beta_2 f(\text{Time}_t) + \beta_3 X_{it} + \varepsilon_{it}$$

In Equation (3), the price is in constant cents (\$2012) and the subscripts i and t denote state and year. The key parameter of interest in Equation (3) is β_1 , which captures the price elasticity of taxable sales. As instrumental variables, I use the combined (state plus federal) tax rate as well as specifications that include these terms separately. The 2SLS estimation allows me to estimate the price elasticity when the pass-through of the tax is not equal to one, which may be the case in

²³ As in the previous section, a key assumption of this approach is that the excise tax rate is exogenous to cigarette consumption. Using a Granger causality test, I do not find evidence that lagged excise rates are substantively or significantly related to consumption after conditioning on prior consumption. These results appear in Appendix Table A4.

this setting as shown in Table 5.²⁴ More generally, the 2SLS specification isolates policy-induced variation in prices that is plausibly exogenous to cigarette demand.

For specifications that use the combined tax rate as an instrument, I include year indicator variables. When using specifications that include separate state and federal tax rates, I use time trends instead of year dummy variables. I estimate specifications with both a common time trend, as well as specifications that include state-specific linear time trends. In X, I control for time-varying state-level variables including state unemployment rate and state population.²⁵

The results suggest that the elasticity of taxable sales of cigarettes ranges from -0.27 to -0.43. Table 6 shows the results. Column (1) shows OLS results, with an estimated elasticity of -0.74. By comparison, the preferred two-stage least squares (2SLS) results in Column (2) that includes flexible time controls (year fixed effects) and the combined tax rate as the instrument suggests an elasticity that is -0.43. Column (3) uses a linear time trend in place of year dummy variables and Column (4) uses this time trend as well as separate federal and state tax variables in the first stage. Compared to Column (2), these results suggest a smaller elasticity. Columns (5) and (6) parallel Columns (3) and (4) but use state-specific linear time trends. These results are also smaller relative to Column (2). Within each set of time controls (linear trend in Columns (3) and (4) and state-specific linear trends in Columns (5) and (6)) the results are similar using the combined rate or separate tax variables in the first stage. Comparing Column (2) to Columns (3) and (5) suggests that much of the difference across specifications is driven by the time controls.

²⁴ Note that the first-stage results in Table 6 do not match the results in Table 5 because Equation (2) is in *levels* whereas Equation (3) is in *logs*.

²⁵ I weight each observation by cigarette sales in order to estimate the national-level elasticity, though the results are similar when I use population-based weights and when I do not use weights.

While the estimated price elasticity of taxable sales is similar to results from Evans, Ringel and Stech (1999) who analyze an earlier period, it is not obvious that consumers would be equally responsive in both periods. In fact, there are several reasons why one might expect greater price sensitivity in the later period. First, the policy induced tax changes that Evans, Ringel and Stech (1999) study do not include federal cigarette excise tax changes with magnitudes similar to the change from CHIPRA. The CHIPRA law changes more than doubled the federal cigarette excise tax and was a relatively larger percentage increase in the price compared to the changes in the earlier period. Second, the CHIPRA law change also introduced large tax differentials across tobacco categories, which could have induced relatively price sensitive cigarette smokers to substitute towards lower-taxed tobacco goods such as pipe tobacco and away from cigarettes. Tax differentials across tobacco products were relatively smaller in the earlier time period. A third reason the elasticity could differ across these studies is the increase in internet commerce relative to earlier years. Results from Goolsbee, Lovenheim and Slemrod (2010) suggest that price sensitive consumers may avoid higher state taxes through internet transactions. The fact that the results here are comparable to Evans, Ringel and Stech (1999) suggests that the price elasticity of taxable sales is relatively stable across tax changes of different magnitudes. For policymakers, this knowledge is potentially important as they analyze the effect of changing tobacco tax rates.

The elasticity results using data on cigarette consumption are consistent with the implied elasticity in cigarette production near the CHIPRA law change, which one would expect if all (or nearly all) produced cigarettes are consumed. Table 7 reports the implied elasticity of production with respect to the retail price change between 2010 and 2009 is roughly -0.4. While the calculations using production data are rough, they do suggest that the implied elasticity of

production is relatively similar to the estimated elasticity of consumption in Table 6. The top panel of Table 7 shows that the percent change in cigarette production for January-March in 2010 and 2009 was roughly 6 percent.²⁶ The middle panel of Table 7 shows that the price change during this period was roughly 18 percent.²⁷

IV. Conclusion

Tobacco excise taxes are one of the principal ways that policymakers can affect tobacco production and consumption decisions. Although the statutory incidence of these taxes is on producers of tobacco products, I find that cigarette excise taxes are more-than passed through to consumers in the form of higher retail prices. The results suggest that consumers respond to this tax-induced price change by reducing consumption. I find that the price elasticity of taxable sales is roughly -0.4. The results also suggest tobacco producers respond to excise taxes. I find strong evidence that manufacturers shift production across substitute goods into lower-taxed products in response to excise tax changes. This shift in production is likely to be reflected in consumption as well, as price-sensitive smokers switch into lower taxed tobacco products.

While the results imply that CHIPRA resulted in meaningful reductions in tobacco production and consumption, the results suggest that federal tax policy could further reduce tobacco consumption by taxing substitute tobacco products equally. Such a shift in policy would also remove tax-induced changes in production across comparable goods. The Tobacco Tax and Equity Act of 2013 proposed by Senator Durbin works towards this goal by taxing pipe and

²⁶ I focus on year-to-date production for the same set of months to remove any seasonal effects. I adjust 2010 production by the trend in annual production between 2007 and 2009 March year-to-date production levels to isolate the quantity change that is likely attributable to the law change.

²⁷ Price data in Orzechowski and Walker (2013) for 2009 include data from the post-CHIPRA period. To estimate prices in 2009 for the months prior to CHIPRA, I inflate 2008 prices by the annual increase in prices from 2007 to 2008.

RYO tobacco at equal rates and by taxing large cigars at a rate that is at least equivalent to the rate of small cigars. This bill also closes the tax differential between smoked tobacco products and smokeless tobacco products.

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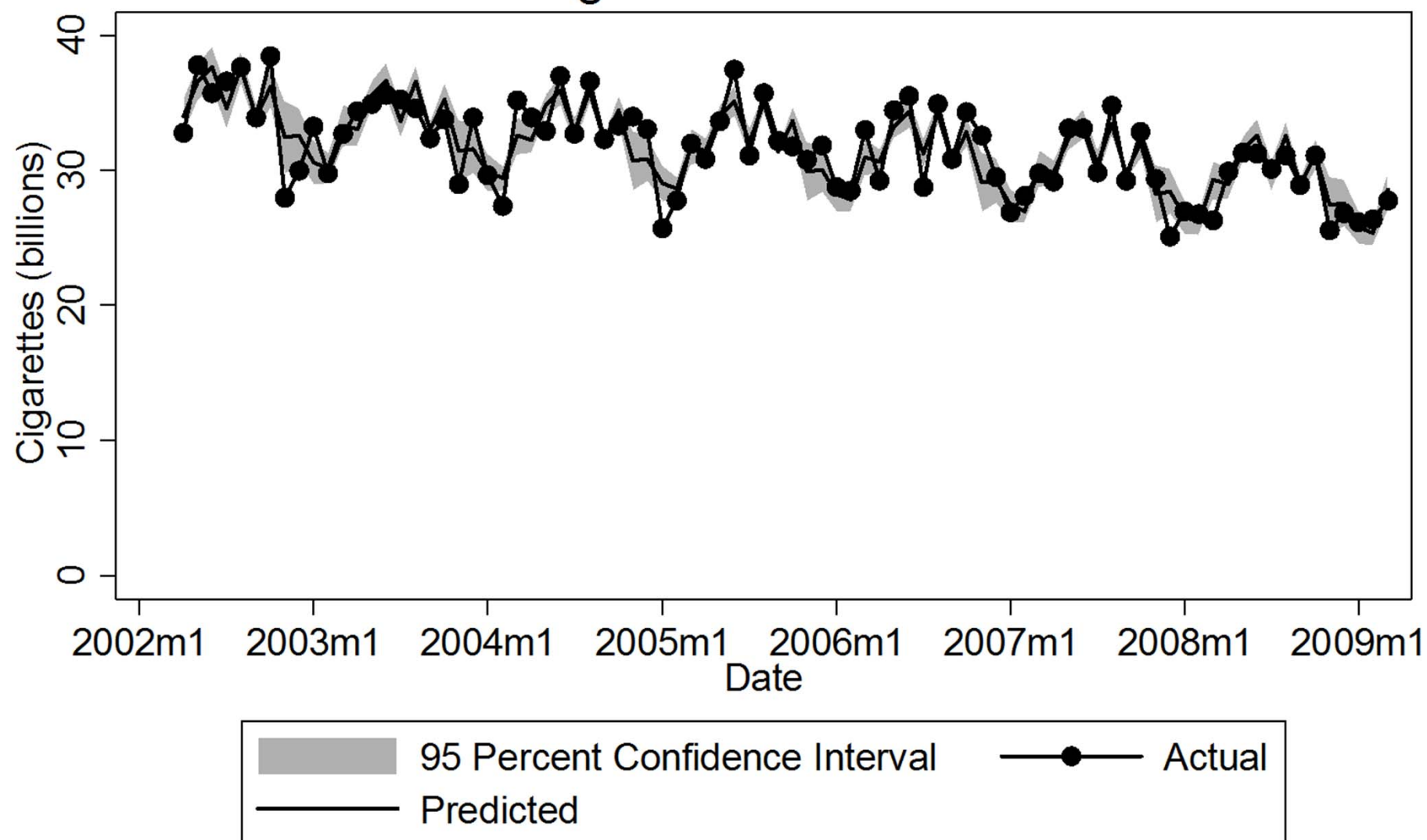
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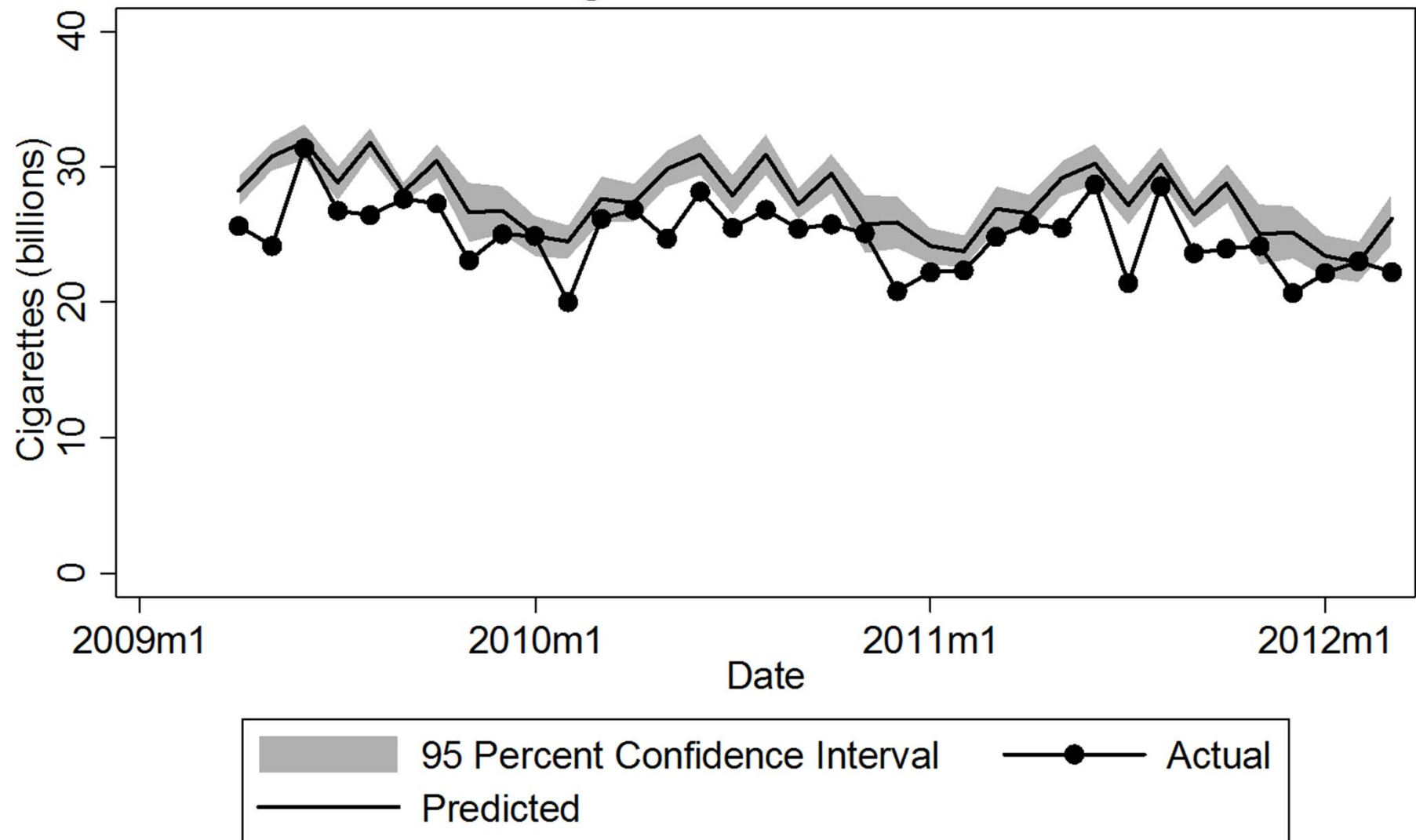
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Figure 1A: In Sample Predicted and Actual
Cigarette Production



Note: Figure shows in sample fit for regression results in Table 3.

Figure 1B: Out of Sample Predicted and Actual Cigarette Production



Note: Figure shows out of sample prediction for regression results in Table 3.

Figure 1C: Out of Sample Predicted Difference
Cigarette Production

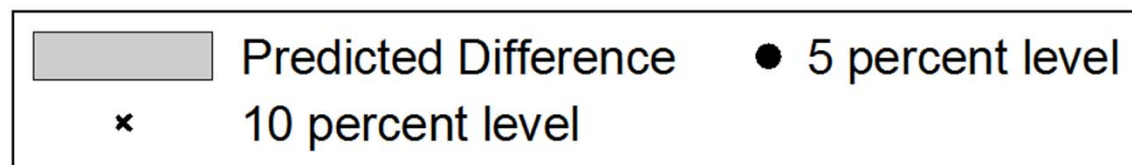
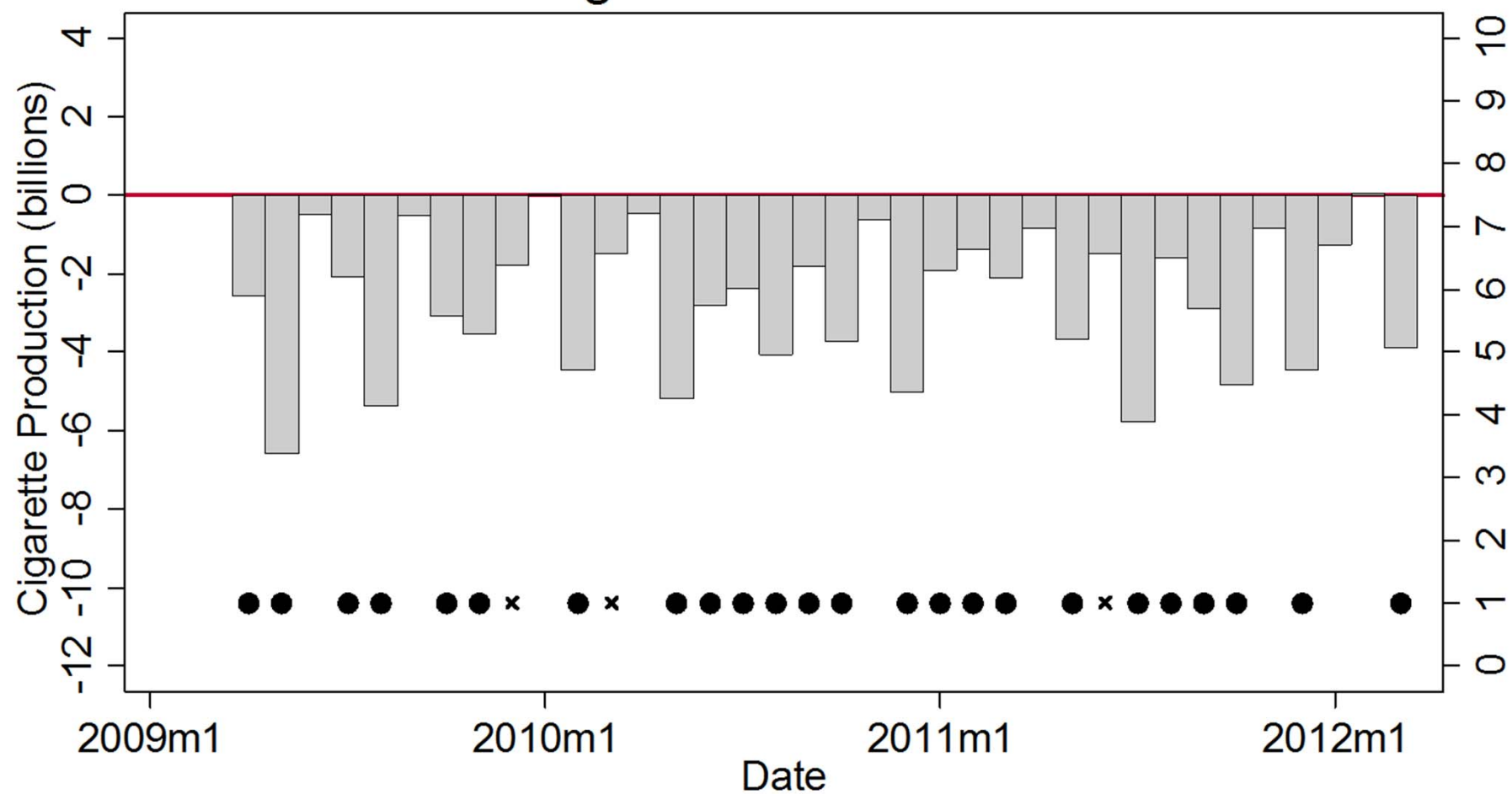
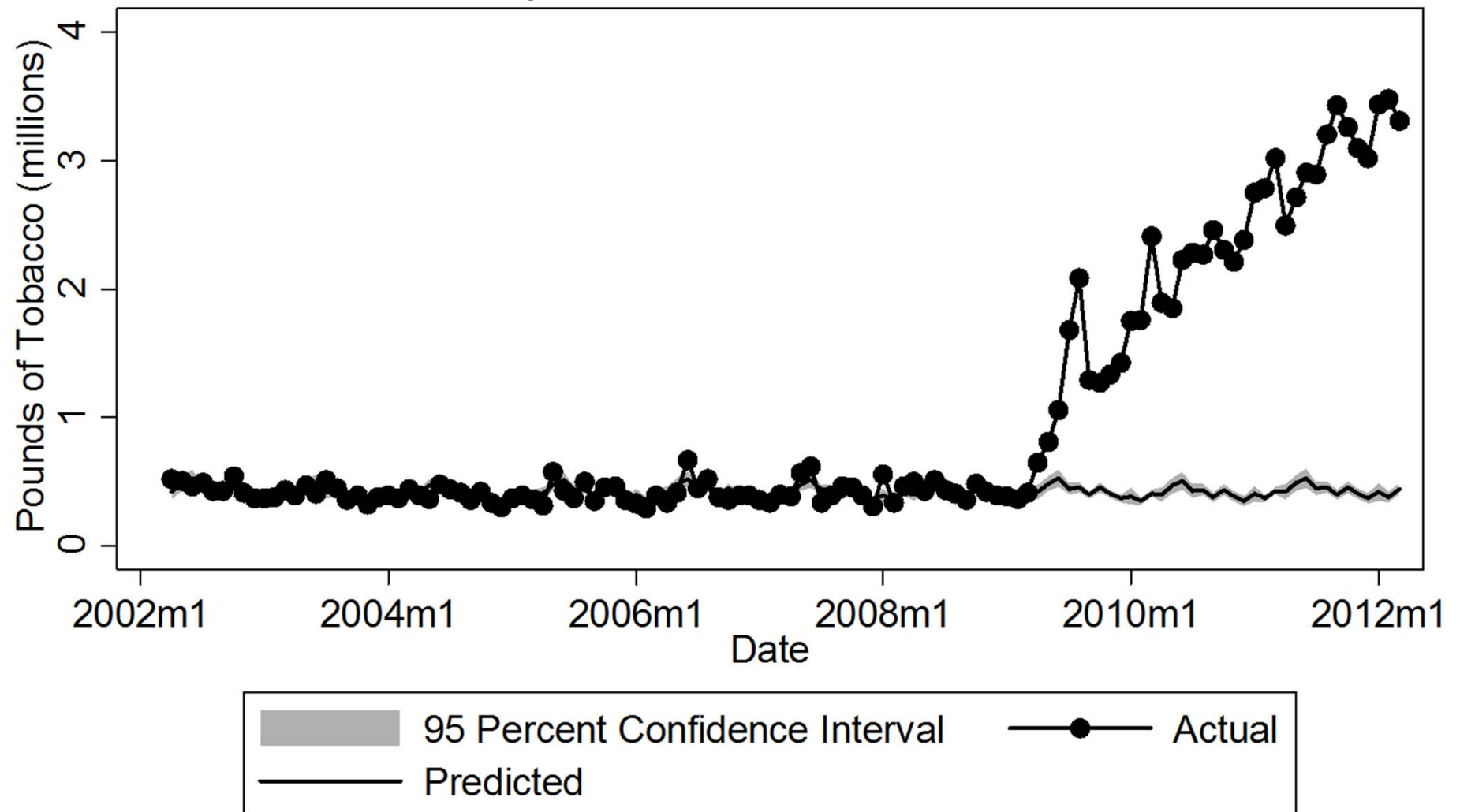
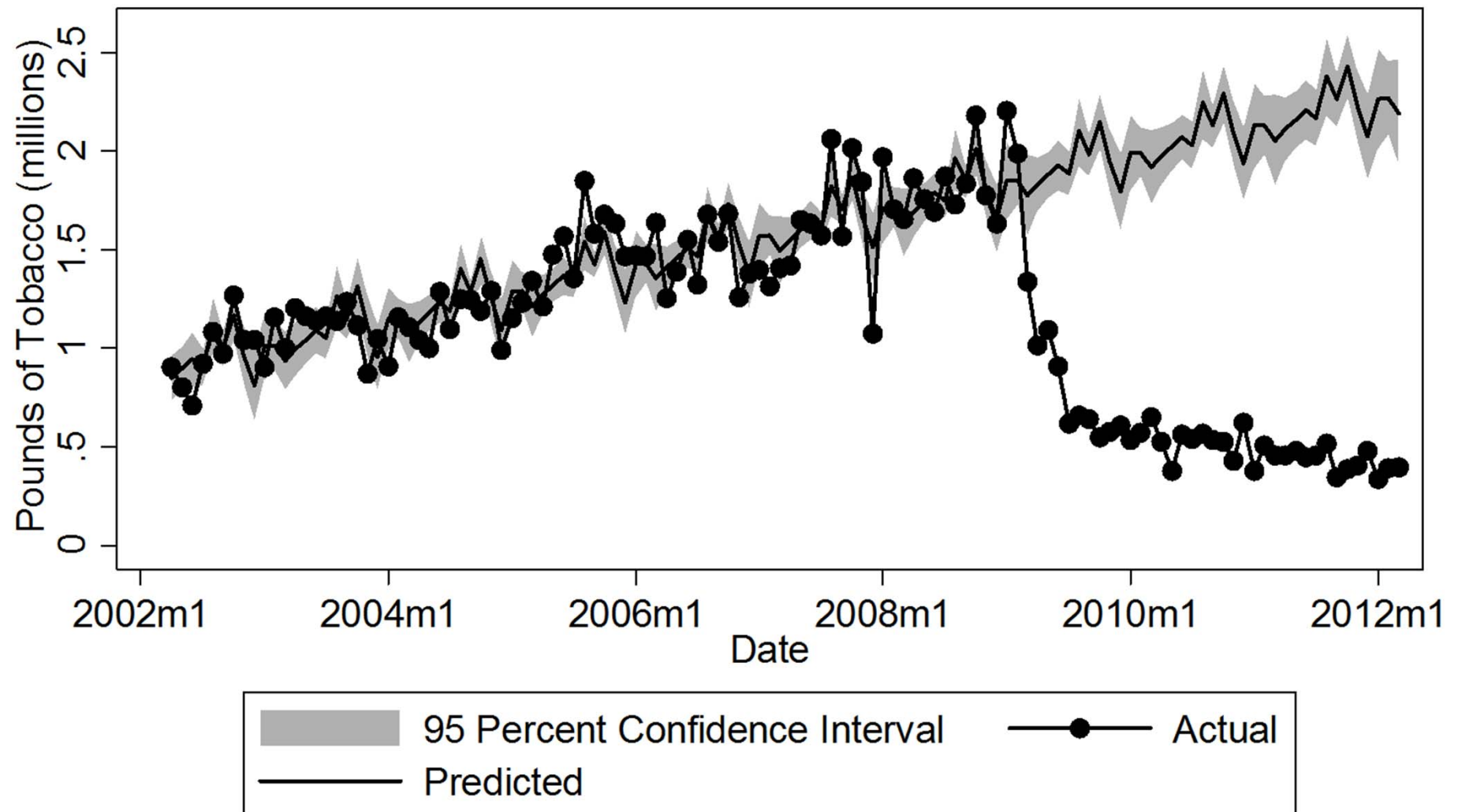


Figure 2: Predicted and Actual
Pipe Tobacco Production



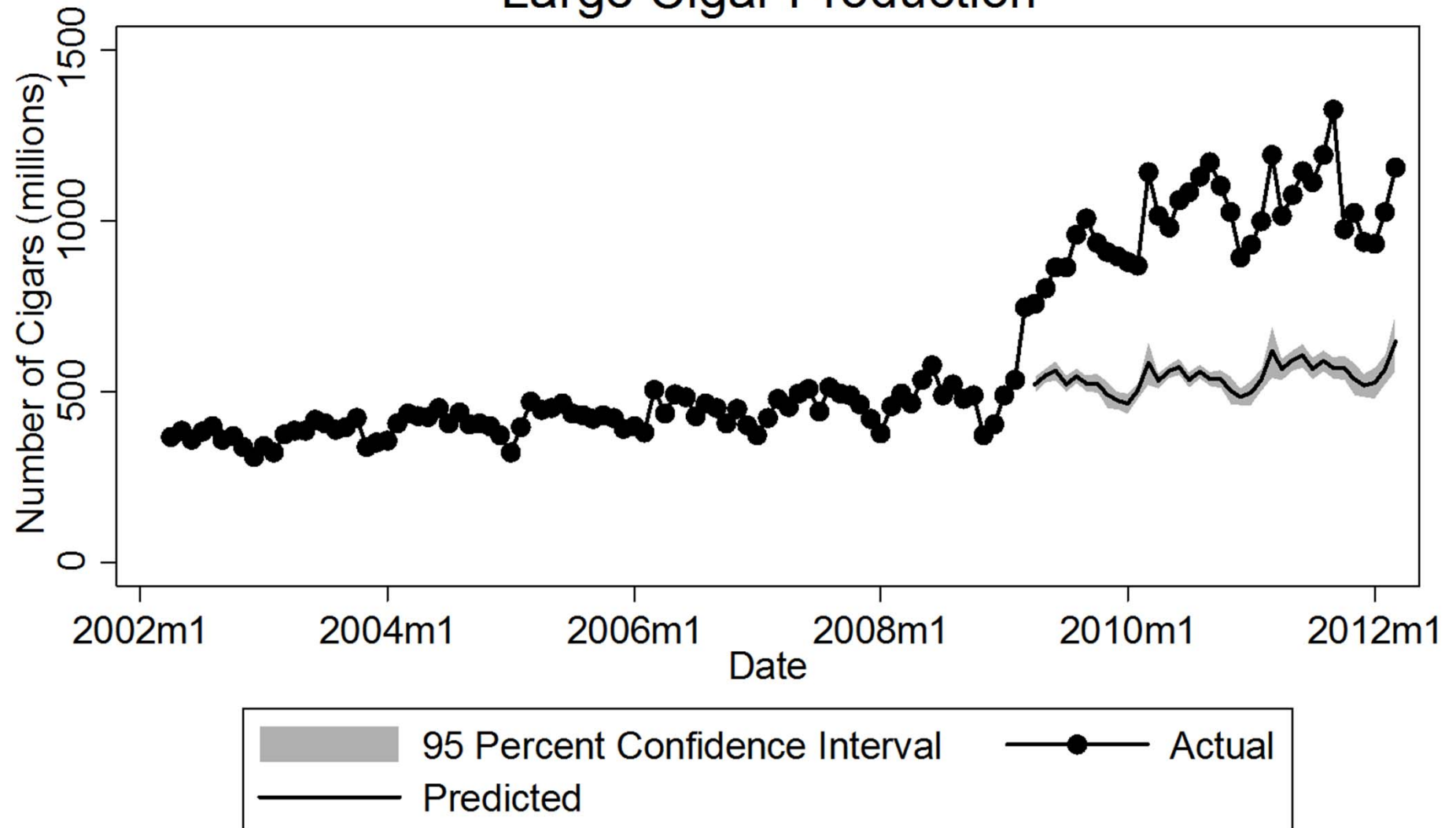
Note: Figure shows in sample fit and out of sample prediction for regression results in Table 3.

Figure 3: Predicted and Actual
RYO Tobacco Production



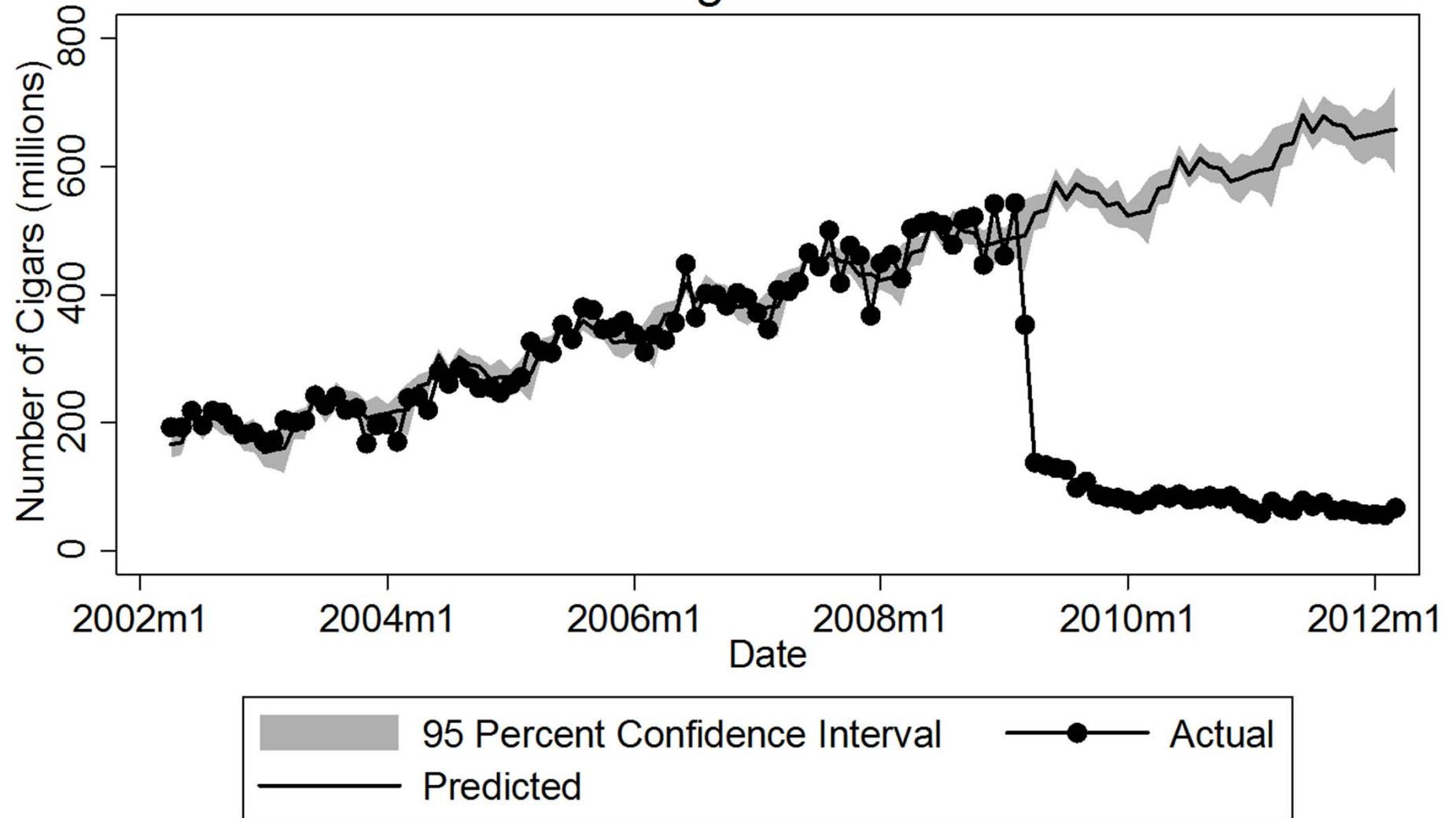
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Figure 4: Predicted and Actual Large Cigar Production



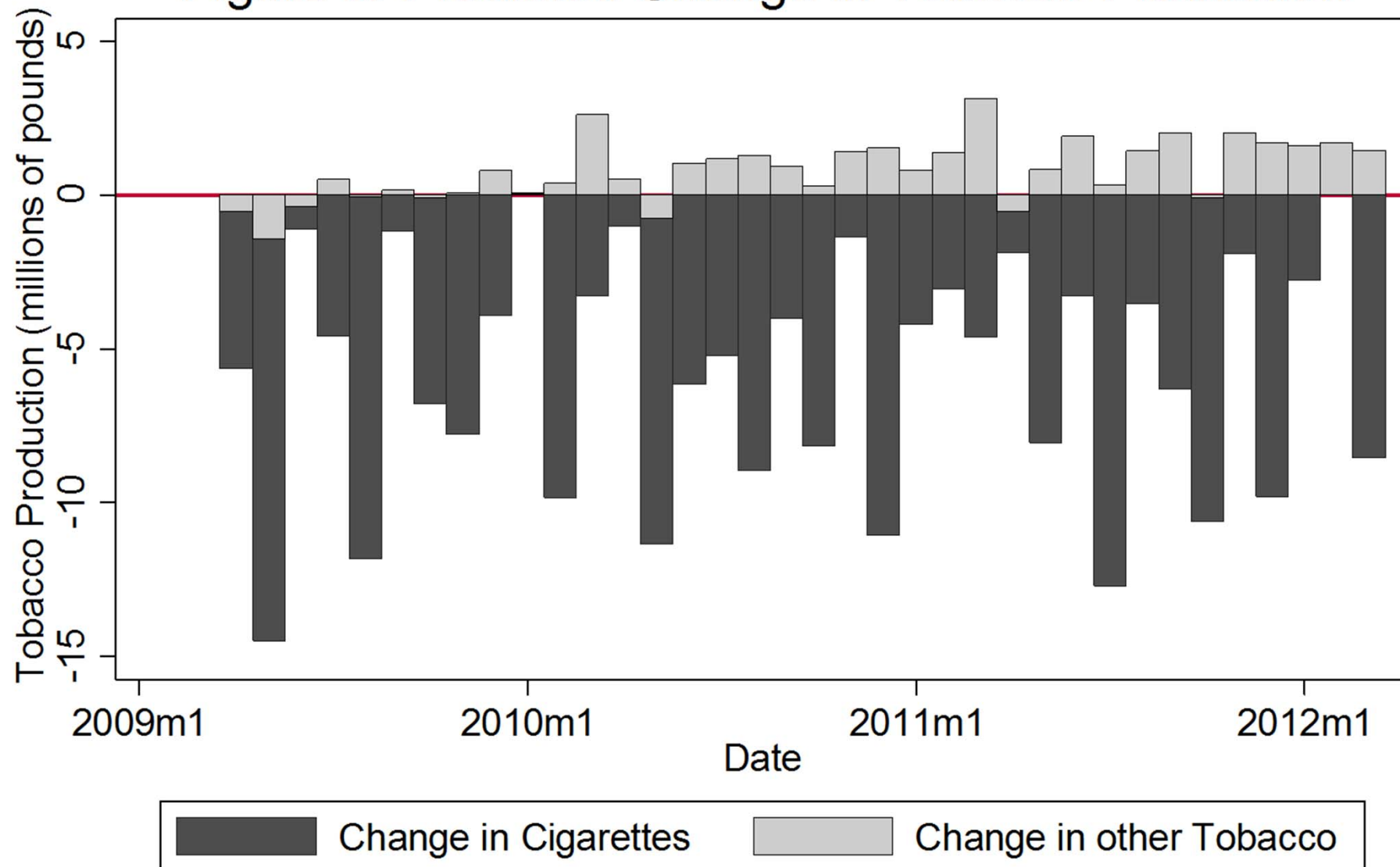
Note: Figure shows in sample fit and out of sample prediction for regression results in Table 3.

Figure 5: Predicted and Actual Small Cigar Production



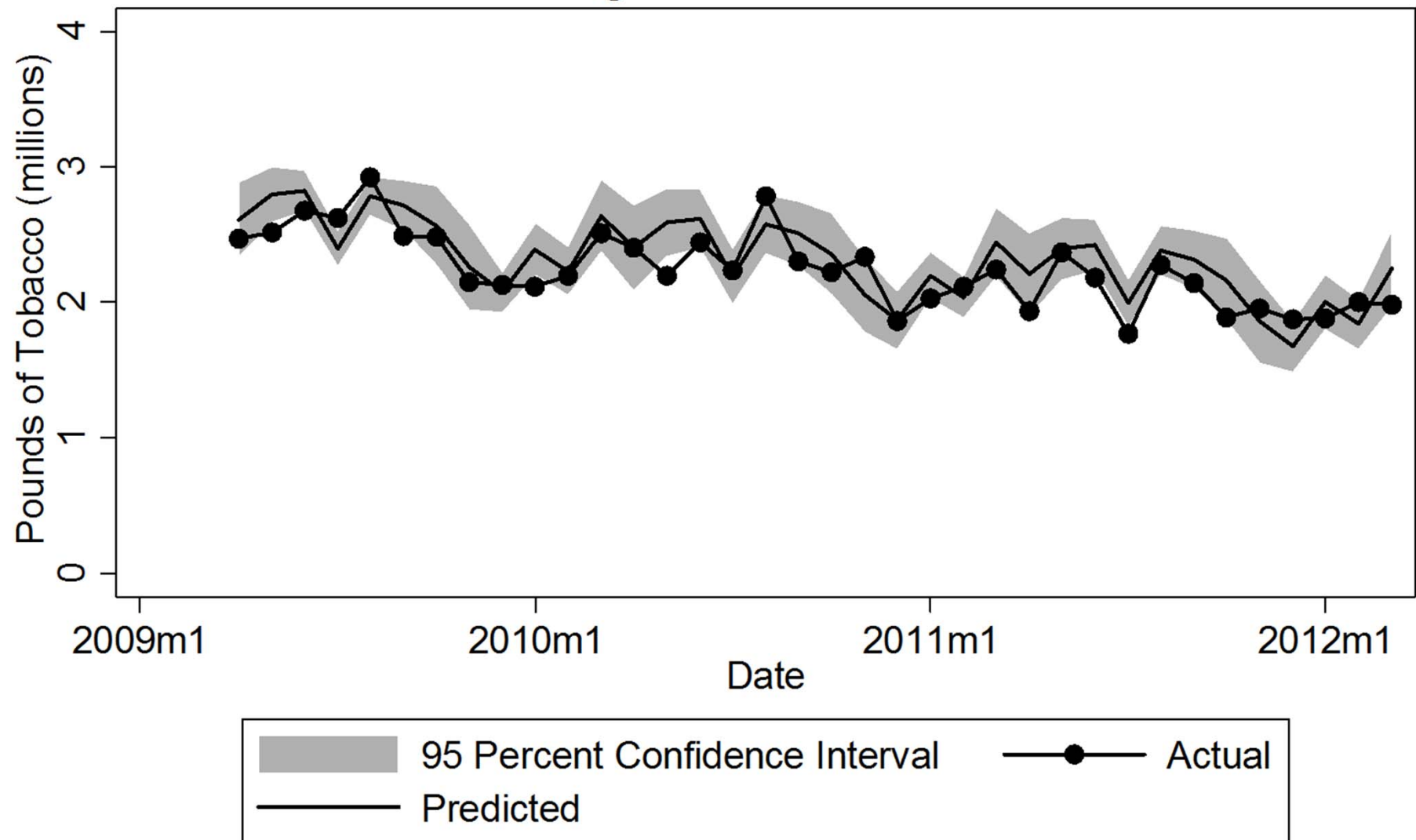
Note: Figure shows in sample fit and out of sample prediction for regression results in Table 3.

Figure 6: Predicted Change in Tobacco Production



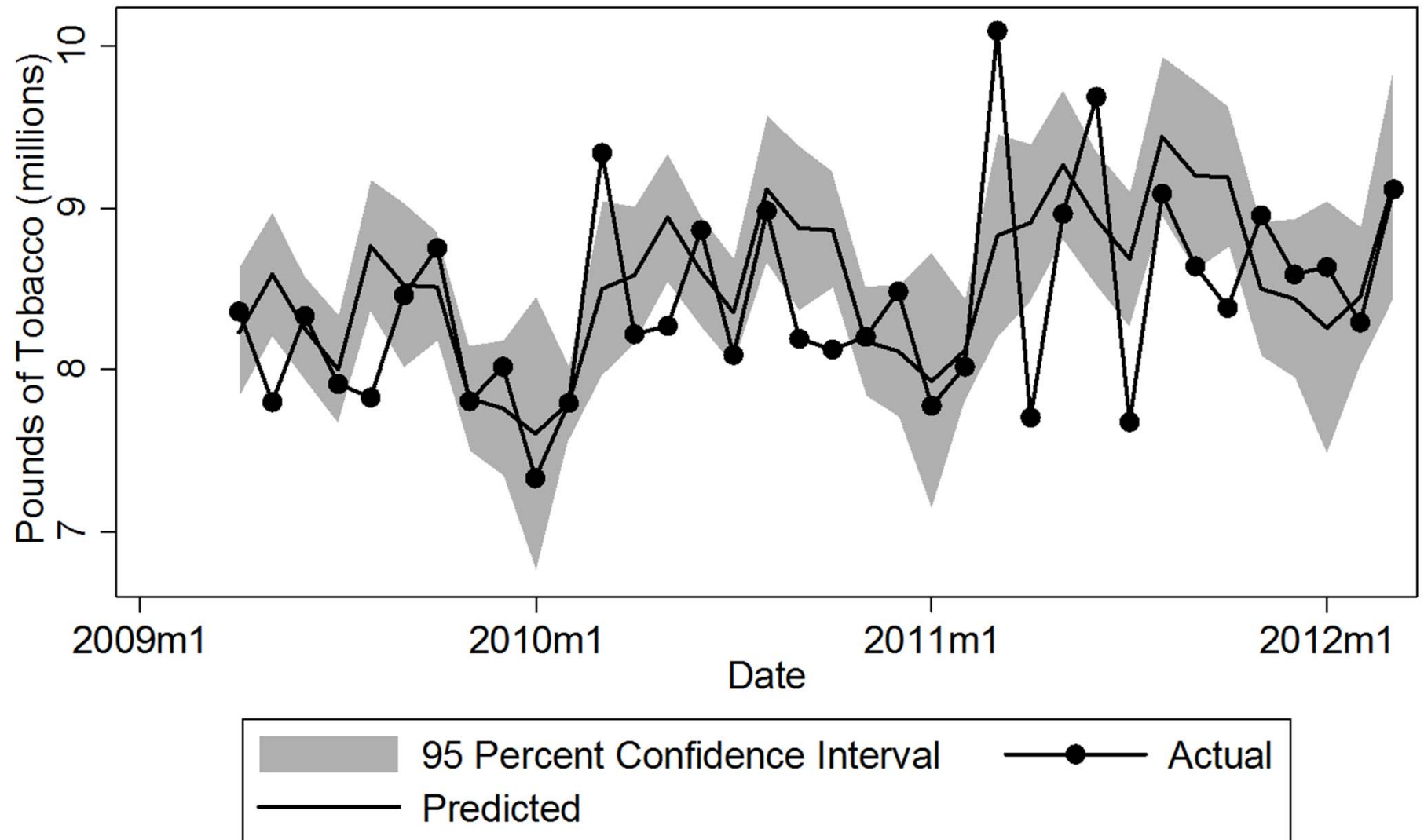
Note: Figure shows net change in pounds for all tobacco products.
See text for conversion from units to pounds.

Figure A1: Out of Sample Predicted and Actual Chewing Tobacco Production



Note: Figure shows out of sample prediction for regression results in Table 3.

Figure A2: Out of Sample Predicted and Actual Snuff Tobacco Production



Note: Figure shows out of sample prediction for regression results in Table 3.

Figure A3: Predicted Change in Tobacco Production

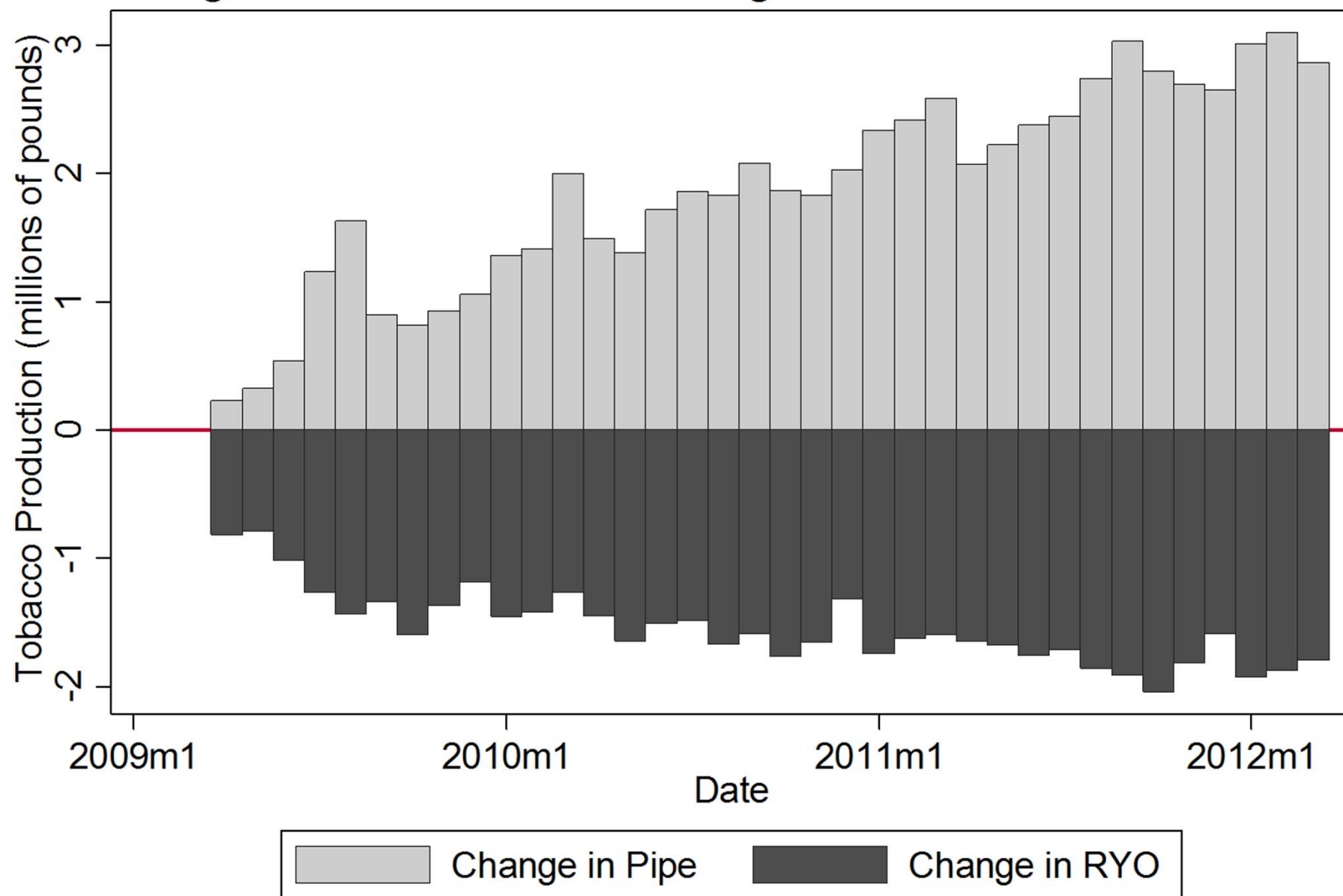


Figure A4: Predicted Change in Cigar Production

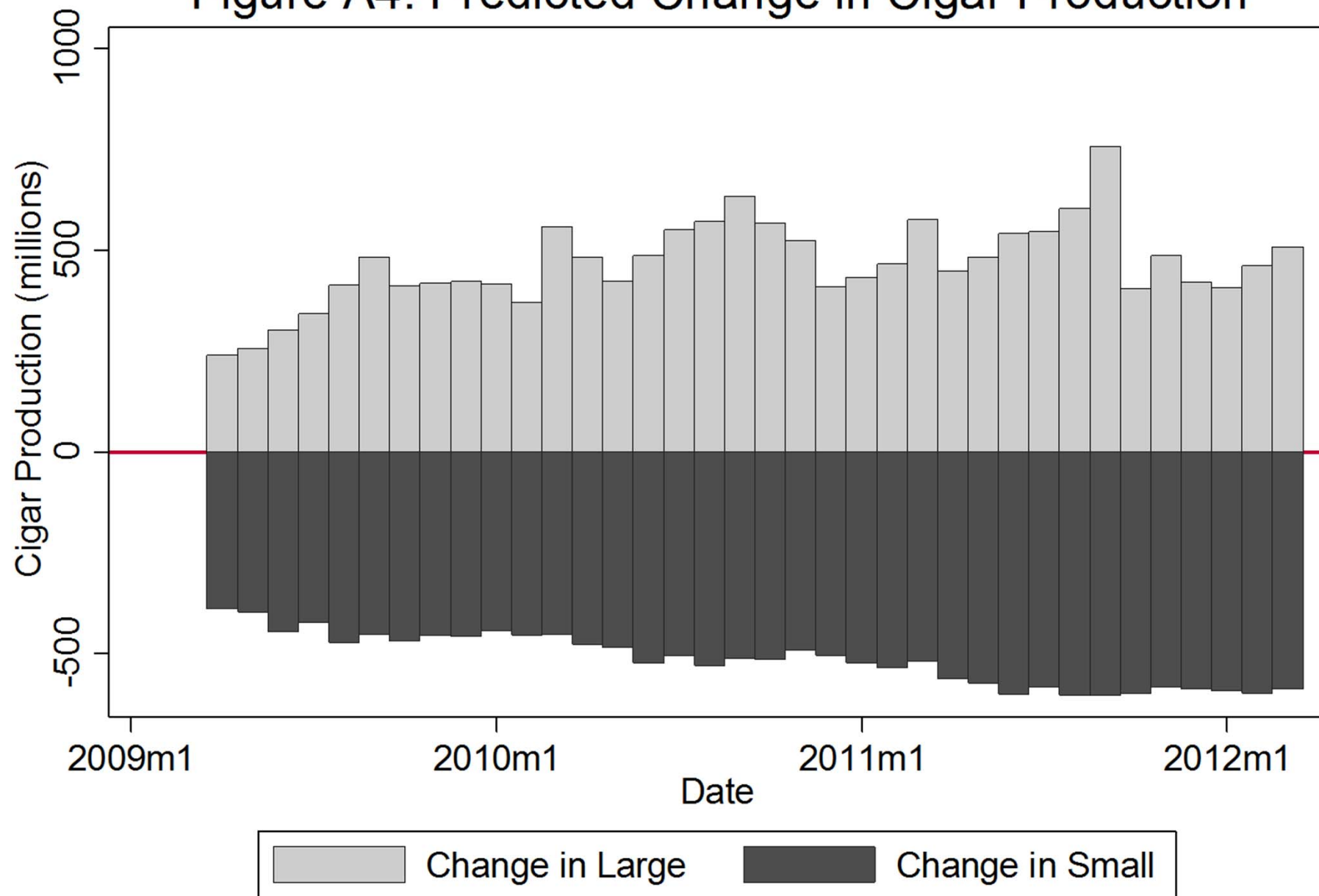


Table 1: Federal Excise Tax Rates for Tobacco Products			
	(1)	(2)	(3)
Effective Dates	January 1, 2000 - December 31, 2001	January 1, 2002-March 31, 2009	April 1, 2009-Today
Cigarettes	\$17.00 per 1,000 equivalent to: \$0.34 per pack	\$19.50 per 1,000 equivalent to: \$0.39 per pack	\$50.33 per 1,000 equivalent to: \$1.0066 per pack
Small Cigars <i>(Weigh 3 lbs. or less per 1,000)</i>	\$1.594 per 1,000	\$1.828 per 1,000	\$50.33 per 1,000
Large Cigars <i>(Weigh more than 3 lbs. per 1,000)</i>	18.063% of sales price with maximum of \$42.50 per 1,000	20.719% of sales price with maximum of \$48.75 per 1,000	52.75% of sales price with maximum of \$402.60 per 1,000
Chewing Tobacco	\$0.17 per pound	\$0.195 per pound	\$0.5033 per pound
Snuff	\$0.51 per pound	\$0.585 per pound	\$1.51 per pound
Pipe tobacco	\$0.9567 per pound	\$1.0969 per pound	\$2.8311 per pound
Roll-your-own tobacco	\$0.9567 per pound	\$1.0969 per pound	\$24.78 per pound
Source: Alcohol and Tobacco Tax and Trade Bureau, U.S. Department of the Treasury.			

Country	Production (kg)	Imports (kg)	Consumption (kg)
USA	15000000	5000000	20000000
Canada	10000000	2000000	12000000
UK	5000000	1000000	6000000
France	3000000	500000	3500000
Germany	2000000	300000	2300000
Italy	1500000	200000	1700000
Spain	1000000	100000	1100000
Japan	500000	1000000	1500000
China	2000000	5000000	7000000
India	1000000	1000000	2000000
Brazil	5000000	500000	5500000
Argentina	3000000	200000	3200000
Australia	2000000	100000	2100000
South Africa	1000000	500000	1500000
South Korea	500000	1000000	1500000
Indonesia	1000000	500000	1500000
Philippines	500000	500000	1000000
Thailand	500000	500000	1000000
Vietnam	500000	500000	1000000
Myanmar	500000	500000	1000000
Laos	500000	500000	1000000
Cambodia	500000	500000	1000000
Malaysia	500000	500000	1000000
Singapore	500000	500000	1000000
Brunei	500000	500000	1000000
Timor-Leste	500000	500000	1000000
East Timor	500000	500000	1000000
West Bank	500000	500000	1000000
Gaza Strip	500000	500000	1000000
Jersey	500000	500000	1000000
Guernsey	500000	500000	1000000
Manx	500000	500000	1000000
Channel Islands	500000	500000	1000000
Faroe Islands	500000	500000	1000000
Greenland	500000	500000	1000000
Åland Islands	500000	500000	1000000
Isle of Man	500000	500000	1000000
Channel Islands	500000	500000	1000000
Faroe Islands	500000	500000	1000000
Åland Islands	500000	500000	1000000
Isle of Man	500000	500000	1000000
Channel Islands	500000	500000	1000000
Faroe Islands	500000	500000	1000000
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Faroe Islands	500000	500000	1000000
Åland Islands	500000	500000	1000000
Isle of Man	500000	500000	1000000
Channel Islands	500000	500000	1000000
Faroe Islands	500000	500000	1000000
Åland Islands	500000	500000	1000000
Isle of Man	500000	500000	1000000
Channel Islands	500000	500000	1000000
Faroe Islands	500000	500000	1000000
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Isle of Man	500000	500000	1000000
Channel Islands	500000	500000	1000000
Faroe Islands	500000	500000	1000000
Åland Islands	500000	500000	1000000
Isle of Man	500000	500000	1000000
Channel Islands	500000	500000	1000000
Faroe Islands	500000	500000	1000000
Åland Islands	500000	500000	1000000
Isle of Man			

	Total Sample		Pre-CHIPRA Sample		Post-CHIPRA Sample		t -stat (3)=(5)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	
Cigarettes	30.51	4.94	31.66	3.29	24.93	2.51	10.98
Pipe Tobacco	1.00	1.00	0.42	0.07	2.31	0.78	22.07
RYO	1.06	0.47	1.39	0.34	0.54	0.17	14.24
Small Cigar	236.43	138.90	332.51	110.75	81.74	21.49	13.45
Large Cigar	581.97	299.86	429.92	64.99	1,011.00	124.98	33.45
Chew Tobacco	3.17	0.80	3.33	0.52	2.24	0.28	11.69
Snuff Tobacco	7.13	1.24	6.85	0.91	8.41	0.60	9.44
Federal Excise Rate (cents/pack)	27.72	40.95	19.50	0	50.33	0	-
State Excise Rate (cents/pack)	14.15	17.74	37.86	8.52	58.73	3.94	14.04
Sample Period	12/2000 to 10/2012		4/2002 to 3/2009		4/2009 to 3/2012		
N	154		84		36		

Notes: Production includes domestic production and imports. Units are billions of cigarettes, millions of cigars, and millions of pounds for all other sources. Source: Alcohol and Tobacco Tax and Trade Bureau, U.S. Department of the Treasury.

Table 3: Counterfactual Production Level Model[illegible]

Table 4: Summary Statistics of Cigarette Prices, Consumption and Excise Tax Rates				
	Include Generic Cigarettes		Exclude Generic Cigarettes	
	(1)	(2)	(3)	(4)
	Mean	Std. Dev.	Mean	Std. Dev.
Price of Pack of Cigarettes (cents)	507.8	114.2	542.50	116.22
Packs of Cigarettes Sold (millions)	348.5	314.4	*	*
Combined Excise Rate (cents/pack)	174.0	59.7	173.9	93.0
Federal Excise Rate (cents/pack)	59.7	23.8	59.7	23.8
State Excise Rate (cents/pack)	114.3	83.3	114.2	83.5
Sample Period	2000 - 2012			
N	663		663	
Notes: * quantity data not available for sales excluding generic cigarettes. All cent amounts are in 2012 cents. Source Orzachowski and Walker (2013).				

Table 5: Tax Incidence Regression Results

	Include Generic Cigarettes					Exclude Generic Cigarettes				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Combined [State + Federal] Tax Rate	1.053 [0.0243]	1.151 [0.0232]		1.167 [0.0333]		1.040 [0.0245]	1.143 [0.0214]		1.169 [0.0312]	
State Tax Rate			1.097 [0.0258]		1.078 [0.0352]			1.081 [0.0236]		1.068 [0.0339]
Federal Tax Rate			1.526 [0.0517]		1.526 [0.0491]			1.576 [0.0469]		1.576 [0.0433]
<i>Time Controls</i>										
Year Dummy Variables	Y	N	N	N	N	Y	N	N	N	N
Trend	N	Y	Y	N	N	N	Y	Y	N	N
State-Specific Linear Trends	N	N	N	Y	Y	N	N	N	Y	Y
Observations	663	663	663	663	663	663	663	663	663	663
Notes: Robust standard errors in brackets. Observations are weighted by number of packs sold at state-year level. All specifications include state unemployment rate and state population.										

Table 6: Elasticity Results

[illegible]

Table 7: Implied Elasticity in Cigarettes Production in Response to CHIPRA

<i>Percent Change Quantity</i>
6.4
<i>Percent Change in Price</i>
18.0
<i>Implied Elasticity</i>
-0.36

Notes: Quantity data are for year to date production measures in March 2010 and March 2009. To adjust for the downward trend in production, I adjust 2010 production by the trend between March year to date in 2008 and March year to date in 2009. To estimate price data in 2009 prior to the law change, I inflate 2008 prices by the change in prices between 2007 and 2008. Data from Orzechowski and Walker, 2013.

Table A1: Granger Causality Test Results for State Tax Rate and Cigarette Production		
	(1)	(2)
Lag[1] Tax Rate	-0.479 [0.0715]	-0.646 [0.0938]
Lag[2] Tax Rate	-0.673 [0.0680]	-0.553 [0.0425]
Lag[3] Tax Rate		-0.243 [0.0602]
Lag[1] Cigarette Production	-0.0114 [0.0607]	-0.0703 [0.0611]
Lag[2] Cigarette Production	0.0780 [0.0495]	0.0857 [0.0490]
Lag[3] Cigarette Production		0.0179 [0.0460]
<i>Test of Lagged Cigarette Production</i>		
F-stat	1.42	1.40
Prob > F	0.245	0.248
Month Dummies	Y	Y
Time Trend	Y	Y
Observations	130	118
Notes: Robust standard errors in brackets. The dependent variable is the current state excise tax rate.		

Table A2: Changes in State Excise Tax Rates for Smoking Tobacco other than Cigarettes, 2008 to 2010					
Rates for 2010					
	Cigarette Rate (\$0.01/pack)	Tax Pipe and RYO same	Rate Large Cigars	Maximum rate small cigars	Change in Pipe, RYO and Cigar Rates from 2008 to 2010
AL	42.5	Yes (max of \$3.52/pound)	\$0.033 to \$40.50 per 1,000	\$4.00 per 1,000	None
AK	200	Yes (same rate as cigars)	75% wholesale price	75% wholesale price	None
AZ	200	Yes (max of \$0.223/oz)	\$71 to \$213 per 1,000	\$20.50 per 1,000	None
AR	115	Yes (same rate as cigars)	68% manufacturer's price	68% manufacturer's price	Increased from 32% to 68% of manufactuer's price
CA	87	Yes (same rate as cigars)	41.10 % of wholesale price	Same as cigarettes (\$43.50 per 1,000)	Decreased from 45.13% to 41.10% of wholesale price
CO	84	Yes (same rate as cigars)	41% of manufacturer's list price	41% of manufacturer's list price	None
CT	300	Yes (same rate as cigars)	20% wholesale price	Same as cigarettes (\$150.00 per 1,000)	None
DE	160	Yes (same rate as cigars)	15% wholesale price	15% wholesale price	None
DC	250	None	None	None	None
FL	133.9	Yes (85% of whoesale price)	None	None	Increased from no tax
GA	37	Yes (10% of retail price)	23% of wholesale price	\$2.50 per 1,000	None
HI	260	Yes (70% wholesale price)	50% of wholesale price	\$110 per 1,000	Increased from 40% of wholesale price
ID	57	Yes (same rate as cigars)	40% of wholesale price	40% of wholesale price	None
IL	98	Yes (same rate as cigars)	18% of wholesale price	18% of wholesale price	None
IN	99.5	Yes (same rate as cigars)	24% of wholesale price	24% of wholesale price	None
IA	136	Yes (50% wholesale price)	50% of wholesale price, capped at \$500 per 1,000	Same as cigarettes (\$68 per 1,000)	None
KS	79	Yes (same rate as cigars)	10% of wholesale price	10% of wholesale price	None
KY	60	Yes (same rate as cigars)	15% of wholesale price	15% of wholesale price	Increased from 7.5% of wholesale price
LA	36	Yes (33% manufactuer's price)	Invoice price greater than (less than) \$120 per \$1,000 taxed at 8% (20%) of price	Invoice price greater than (less than) \$120 per \$1,000 taxed at 8% (20%) of price	None
ME	200	Yes (same rate as cigars)	20% of wholesale price	20% of wholesale price	None
MD	200	Yes (same rate as cigars)	15% of manufacturer's list price	15% of manufacturer's list price	None
MA	251	Yes (same rate as cigars)	30% of price paid by retail vendor	Same as cigarettes (\$125.50 per 1,000)	None
MI	200	Yes (same rate as cigars)	32% of wholesale price	32% of wholesale price	None
MN	156	Yes (same rate as cigars)	70% of wholesale price	70% of wholesale price	None
MS	68	Yes (same rate as cigars)	15% of manufacturer's list price	15% of manufacturer's list price	None
MO	17	Yes (same rate as cigars)	10% of manufacturer's list price	10% of manufacturer's list price	None
MT	170	Yes (same rate as cigars)	50% of wholesale price	Same as cigarettes (\$85 per 1,000)	None
NE	64	Yes (same rate as cigars)	20% of wholesale price	20% of wholesale price	None
NV	80	Yes (same rate as cigars)	30% of wholesale price	30% of wholesale price	None
NH	178	Yes (same rate as cigars)	48.59% of wholesale price	Same as cigarettes (\$89 per 1,000)	Pipe and RYO increased from 19% of wholesale price, cigars were not previously taxed.

NJ	270	Yes (same rate as cigars)	30% of wholesale price	30% of wholesale price	None
NM	91	Yes (same rate as cigars)	25% of wholesale price	25% of wholesale price, although some taxed the same as cigarettes	None
NY	275	Yes (same rate as cigars)	46% of wholesale price	46% of wholesale price	Increased from 37% of wholesale price
NC	45	Yes (same rate as cigars)	10% of wholesale price	10% of wholesale price	None
ND		No (Pipe taxed at 28% of wholesale price, RYO at \$0.022 per 0.9 oz)	28% of wholesale price	28% of wholesale price	
	44				None
OH	125	Yes (same rate as cigars)	17% of wholesale price	17% of wholesale price	None
OK	103	Yes (80% of list price)	\$110-\$120 per 1,000	\$36 per 1,000	None
OR	118	Yes (65% of wholesale price)	65% of wholesale price, capped at \$500 per 1,000	65% of wholesale price, capped at \$500 per 1,000	None
PA	160	None	None	None	None
RI	346	Yes (40% of wholesale price)	40% of wholesale price, capped at \$500 per 1,000	Same as cigarettes (\$173 per 1,000)	None
SC	7	Yes (same rate as cigars)	5% of manufacturer's price	5% of manufacturer's price	None
SD	153	Yes (same rate as cigars)	35% of wholesale price	35% of wholesale price	None
TN	62	Yes (6.6% of wholesale price)	6.6% of wholesale price	Same as cigarettes (\$31 per 1,000)	None
TX	141	Yes (40% of manufacturer's price)	\$7.50 to \$15.00 per 1,000	\$10 per 1,000	None
UT	69.5	Yes (same rate as cigars)	35% of manufacturer's list price	35% of manufacturer's list price	None
VT		No (pipe taxed at 92% of wholesale price, RYO at \$19.91/pound)	92% of wholesale price	Same as cigarettes (\$112 per 1,000)	Pipe and large cigars increased from 41% of wholesale price, RYO from \$15.91, small cigars from \$89.50 per 1,000)
	224				
VA	30	Yes (same rate as cigars)	10% of manufacturer's price	10% of manufacturer's price	None
WA	302.5	Yes (75% of wholesale price)	75% of wholesale price, capped at \$500 per 1,000	75% of wholesale price, capped at \$500 per 1,000	None
WV	55	Yes (same rate as cigars)	7% of wholesale price	7% of wholesale price	None
WI	252	Yes (71% of manufacturer's invoice price)	71% of manufacturer's invoice price, capped at \$500 per 1,000	71% of manufacturer's invoice price, capped at \$500 per 1,000	None
WY	60	Yes (same rate as cigars)	20% of wholesale price	20% of wholesale price	None

Notes: For ease of comparison, when possible units are standardized to pounds or 1,000s of cigars.

Source: Orzachowski and Walker 2007 and 2009, Tables 12B

Table A3: In-Sample Regression Results for Alternate Sample Dates

[illegible]

Table A4: Granger Causality Test Results for Combined Tax Rate

	(1)	(2)	(3)	(4)	(5)	(6)
Lag[1] Tax Rate	0.917 [0.157]	0.890 [0.139]	0.699 [0.0764]	0.676 [0.0652]	0.472 [0.230]	0.876 [0.146]
Lag[2] Tax Rate	0.101 [0.208]	0.000135 [0.138]	0.144 [0.147]	-0.0522 [0.0646]	0.0389 [0.156]	0.00198 [0.144]
Lag[3] Tax Rate	-0.130 [0.176]		-0.246 [0.137]		-0.0288 [0.134]	
Lag[1] Price	-0.234 [0.167]	-0.213 [0.141]			0.218 [0.214]	-0.209 [0.150]
Lag[2] Price	-0.0293 [0.153]	-0.0773 [0.118]			-0.0168 [0.144]	-0.0409 [0.126]
Lag[3] Price	-0.0713 [0.144]				-0.124 [0.123]	
Lag[1] Quantity			0.0576 [0.0350]	0.00858 [0.0256]	0.0286 [0.0323]	0.0117 [0.0250]
Lag[2] Quantity			0.0164 [0.0360]	0.0476 [0.0271]	0.0413 [0.0351]	0.0382 [0.0282]
Lag[3] Quantity			-0.0607 [0.0426]		-0.0710 [0.0398]	
<i>Test of Lagged Prices</i>						
F-stat	1.430	2.600			0.990	1.660
Prob > F	0.232	0.075			0.399	0.191
<i>Test of Lagged Quantities</i>						
F-stat			1.330	2.270	1.450	1.540
Prob > F			0.264	0.104	0.227	0.216
State Dummies	Y	Y	Y	Y	Y	Y
Year Dummies	Y	Y	Y	Y	Y	Y
Observations	459	510	459	510	459	510
R-squared	0.931	0.927	0.931	0.928	0.932	0.928

Notes: Robust standard errors clustered at the state level in brackets. The dependent variable in all specifications is the current state tax rate.